

AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL

Submission: model_3

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

The solution correctly proves that the a_i are congruent modulo g , meeting the partial-progress milestone. Beyond that the upper-bound argument is fundamentally flawed: it never addresses $k \geq 4$ (**NEC3**), it introduces an unexplained and incorrect bound $g < 5 \times 10^5$, and it then only tests examples over a limited range, arriving at the *wrong* answer (502945 instead of 998285). It therefore scores **1** for the common-residue milestone only.